



UNIT 13 REAL ANALYSIS

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13.0 OBJECTIVES

After going through this unit, you should be able to:

- explain the essence of the discipline of real analysis;
- explain the concepts of a sequence and related concepts like term, length of sequence, and subsequence;
- define well-known types of sequence such as Arithmetic Progression (A.P.), Geometric Progression (G.P.), (H.P.) and use these in problem-solving;
- define and characterise sequences with different properties like boundedness, Monotonicity, Convergence, Divergence, Oscillation;
- define the concepts of limit and limit point of a sequence;

- define, explain and, use in economic problem solving, the Bolzano-Weierstrass theorem for sequences as well as for sets;
- define some specific types of (sub-)sets of real numbers such as Open Sets, Closed Sets, Bounded Sets, Compact Sets; and
- apply the concepts of function, limit, and continuity etc. *in respect of several variables over $\mathbb{R}^n$*

13.1 INTRODUCTION

This unit will cover real analysis in detail but to start with we will first understand a few basics about real analysis, like what is real analysis, difference between calculus and real analysis, requirement of real analysis by calculus and the areas where real analysis is applied.

- a) **Real analysis**, and in general, **Mathematical Analysis**, are mathematical disciplines, which dominantly use **analytical skills** for solving mathematical problems. Real analysis is a branch of Mathematical Analysis, that restricts itself to the domain of real numbers. **Analytical skills** include the human capability to *recursively*
- relate a hard problem to relevant simpler problems, and/or break down it into smaller sub problems which are already solved or are easily solvable,
 - solve independently the simpler problems, or smaller sub problems; and then
 - reassemble/ integrate the solutions of the simpler problems, and/or sub problems into a solution of the (difficult) problem.

For example,

- Finding the behaviour of the members of the following series as we proceed further

from the first term

$$1, (1/2), (1/2)^2, (1/2)^3, (1/2)^4, \dots,$$

through examining the behaviour/ pattern of some of the initial terms like 1, $1/2$, $(1/2)^2$, i.e. through examining only some of the initial terms,

- Finding the sum of a series like

$$S = 1 + (1/2) + (1/2)^2 + (1/2)^3 + (1/2)^4 + \dots$$

through examining partial sums

$$S_1 = 1, \quad S_2 = 1 + (1/2) = 3/2, \quad S_3 = 1 + (1/2) + (1/4) = 7/4, \dots$$

- *Calculating change at a particular point* of time/space is quite a *difficult* problem, which is attempted to be solved *through evaluating* change over *small intervals containing the point*, and, gradually making the intervals smaller so as to coincide with the point. *Evaluating change over an interval is simpler.*

- Finding area bounded by an irregular figure by approximating it through the areas of small 2-dimensional closed *regular* figures, where the sizes of the closed figures is gradually reduced.
- Developing recursively theories of larger sets of numbers from those of the smaller sets: The actual constructions and theories of reals from those of rationals; that of rationals from those of integers, and so on.

Some of the categories of analytical skills include logical reasoning, critical thinking, and data analysis.

b) **Subject matter of Real Analysis:** It is the study of *behaviour and properties* of real numbers (R), of sequences and series of real numbers, and of *real functions*. Most of the functions studied are *real-valued*, however, the codomain may be any other non-empty set also. A function is a

- *real function* if its *domain* is R, or subset of R containing an interval.
- *real-valued* if its *codomain* is R.

Limit is the fundamental concept of analysis both for real sequences/series and for real-valued functions. For *sequences and series*, the other fundamental concepts include *convergence*. For *functions*, other fundamental concepts include *continuity, differentiability and integrability*.

c) **Difference between Calculus and Real Analysis:** Despite of the fact that subject matter of each of calculus and real analysis involves most of the same concepts including limit, continuity, derivative, integrals etc., the two disciplines approach these topics from different perspectives: Real analysis provides sound theoretical foundations and justifications for the calculations done through calculus. For example, *in calculus*, we apply the rule of finding derivative of quotient u/v , of two derivable functions u and v of x , with $v \neq 0$, using the Quotient Rule:

$$d(u/v)/dx = (v \, du/dx - u \, dv/dx)/v^2 \quad \dots (13.1)$$

However, the *justification* for the facts that—(i) $d(u/v)/dx$ exists, and (ii) Equation (A) above is true—is provided by *real analysis*.

Similarly, the fundamental theorem of calculus is used for finding definite integral of a function f for which an anti-derivative F is known. However, the *proof* of fundamental theorem is subject matter of *Real Analysis*.

d) **Calculus requires the theoretical foundations from Real Analysis:** Let us illustrate this through a couple of examples:

Example 13.1

Find the sum of the series $1 + (1/2) + (1/2)^2 + (1/2)^3 + \dots$,

For this purpose, consider the following method of finding sum of the series:

Let $S = 1 + (1/2) + (1/2)^2 + (1/2)^3 + \dots$

Then $2S = 2 + [1 + (1/2) + (1/2)^2 + (1/2)^3 + \dots]$

$$= 2 + S$$

$$\text{Therefore } 2S - S = 2.$$

$$S = 2$$

is the required solution.

Apply the above method to find sum of the following series

$$S' = 1 + 2 + 2^2 + 2^3 + \dots$$

Then, through the above method, we get

$$2S' = 2[1 + 2 + 2^2 + 2^3 + \dots] = 2 + 2^2 + 2^3 + \dots = S' - 1$$

Therefore $S' = -1$, which is *completely absurd*.

Reason for absurdity: The method is valid only under certain conditions. Determining validity of conditions is the concern of Theory (in our case, *Real Analysis*). The role of calculus is to apply the method, under the validity conditions determined by Real Analysis.

Example 13.2

$$\text{Let } L = \lim_{m \rightarrow \infty} (y)^m$$

$$\text{Further, by letting } m = n+1, \text{ we get } L = \lim_{n+1 \rightarrow \infty} (y)^{n+1} = \lim_{n+1 \rightarrow \infty} y (y)^n = y \lim_{n+1 \rightarrow \infty} (y)^n$$

But as $n+1 \rightarrow \infty$ then $n \rightarrow \infty$, therefore,

$$L = \lim_{n+1 \rightarrow \infty} (y)^{n+1} = \lim_{n+1 \rightarrow \infty} y (y)^n = y \lim_{n+1 \rightarrow \infty} (y)^n = y \lim_{n \rightarrow \infty} (y)^n = y(L).$$

$$(y - 1)L = 0, \text{ implying either } y=1, \text{ or, } L=0$$

$$\text{Thus, for all } y\text{'s with } y \neq 1, \text{ we have } \lim_{m \rightarrow \infty} (y)^m = L = 0.$$

But again, *this is absurd*, because by taking $y = 2$, we get the limit of the sequence $2, (2)^2, (2)^3, (2)^4, \dots$ as **zero**. Again, determining/ explaining the absurdity is the job of theory (Real Analysis). The role of calculus is to apply the method under the conditions determined by the theory (Real Analysis).

- e) **Application Domains:** In the area of natural sciences, social sciences and engineering, the notion of *continuous change* plays a significant role. Specifically, in economics, some of the areas of its applications include expected utility theory, welfare economics, financial economics, cooperative games, e. g. the Nash Bargaining Solution.

13.2 SEQUENCES

We have an *intuitive* idea of what a sequence is, e. g. a sequence of events in a cultural program or we have an idea that $(1, 2, 4, 8, \dots)$ is a sequence of numbers which are powers of 2. Sequence is a concept in which entities of its concern come, in some order, one after another. A sequence, may be considered either as a (special type) of set or as a (special type) of function.

Notation: Before we proceed further, it may be noted that the (standard) notation for a sequence is the same as that for an ordered set, e. g., $(1, 2, 4, 8, \dots)$, which is bounded by parentheses ‘()’ instead of the set notation $\{1, 2, 4, 8, \dots\}$, bounded by braces ‘{}’.

13.2.1 Set-type Conceptualization of a Sequence

A sequence is a sort of a set, such that it

- **is an ordered set:** (E, V, I, L) and (V, I, L, E) are *distinct sequences* formed out of the *same set* $\{E, I, V, L\}$.
- **is an enumerated set,** i.e., all the elements can be listed one after another according to some ordering. For example, in the sequence $\{0, 1, 4, 9, \dots\}$ of squares of integers, 1 is listed after 0; 4 is listed after 1, and so on. This sort of listing is *not possible* for the elements of the *set of real numbers* with any ordering. This sort of listing is *not possible* even for the elements of a small interval like $[0, 1]$. For example, w.r.t the order-relation ‘<’, for the purpose of listing, we do not know the real number immediately next to 0. Thus, any interval of real numbers cannot be represented as a sequence w.r.t the relation ‘<’. Actually, $\mathbb{R}$, or any of its sub-interval, cannot be represented by a sequence w.r.t *any* relation, because $|\mathbb{R}| > |\mathbb{N}|$, where $|S|$ denotes the number of elements in (or cardinality of) the set S .
- **unlike a set, allows repetition of its elements,** e. g. $(1, 2, 4, 2, 4)$ is a correct sequence representation, though $\{1, 2, 4, 2, 4\}$ is not accepted as proper notation for the set $\{1, 2, 4\}$.

Alternatively, more formally, a sequence is considered as a function as follows.

13.2.2 Sequence as a Function

A *sequence* of real numbers is a function

$$f: S \rightarrow \mathbb{R},$$

where

- S , the **domain**, is either
 - the set $\mathbb{N}$ of natural numbers, or
 - an interval subset of $\mathbb{N}$ —e. g. (i) integers in the interval $[10, 20]$, or in the interval $[10, \infty[$, or even
 - integers in the finite extensions $[-k, n]$, $[-k, \infty[$, for $k \in \mathbb{N}$ & $n \in \mathbb{N}$, e. g. integers in $[-10, 20]$, or in $[-10, \infty[$, and
- $\mathbb{R}$, the **codomain**, is the set of real numbers. The codomain may be $\mathbb{Q}$, the set of rational numbers; $\mathbb{Z}$, the set of integers; $\mathbb{C}$, the set of complex numbers; or, in general any non-empty set.

According to this definition of the sequence, our main concern is the images of the function, written in order from the first towards the last, e. g., for the sequence

$$f: \mathbb{N} \rightarrow \mathbb{R},$$

we are interested in the ordered set $(f(1), f(2), f(3), \dots)$,

which, most of the time, we write as $(f_1, f_2, f_3, \dots)$.

A shorthand for $(f_1, f_2, f_3, \dots)$ is (f_n) , $n = 1, 2, 3, \dots$, or even $(f_n)_{n \in \mathbb{N}}$

Examples of sequence as a function

Example 13.3

The sequence of all non-negative integral powers of 2, may be considered as the function

$$\text{ISQ}: \mathbb{N} \rightarrow \mathbb{N}, \text{ s. t. } \quad (\text{ISQ, our shorthand for Integer Square})$$

$$\text{ISQ}(n) = n^2 \text{ for all } n \in \mathbb{N}.$$

But our interest here is in the set of images

$$(\text{ISQ}(1), \text{ISQ}(2), \text{ISQ}(3), \dots),$$

Which, in context of sequences, is written as

$$(\text{ISQ}_1, \text{ISQ}_2, \text{ISQ}_3, \dots) \text{ with } \text{ISQ}_n = 2^n,$$

$$\text{or, } \text{ISQ}_n = 2^n, n = 1, 2, 3, \dots \text{ or, } (2^n)_{n \in \mathbb{N}}$$

Example 13.4

However, if we are interested only in the sequence of powers of 2, which are two-digit integers, then the functional representation of the sequence may be

$$\text{ISQ}: [4, 9] \rightarrow \mathbb{N}, \text{ s. t. } \text{ISQ}(n) = n^2 \text{ for all } n.$$

And, then our interest is in the set of images

$$(\text{ISQ}_4, \text{ISQ}_5, \dots, \text{ISQ}_9) = (2^4, 2^5, 2^6, 2^7, 2^8, 2^9)$$

Which in shorthand is written as

$$\text{ISQ}_n = 2^n, \text{ for } n = 4, 5, \dots, 9, \text{ or}$$

$$\text{ISQ}_n = (2^n)_{4 \leq n \leq 9}, n \in \mathbb{N} \text{ may be taken as understood}$$

Example 13.5

Further, if we are interested only in the sequence of powers of 2, with values between 2^{-10} to 2^{10} , then the functional representation of the sequence may be

$$\text{ISQ}: [-10, 10] \rightarrow \mathbb{N}, \text{ s. t. } \text{ISQ}(n) = n^2 \text{ for all } n \in [-10, 10].$$

And, then our interest is in the set of images

$$(\text{ISQ}_{-10}, \text{ISQ}_{-9}, \dots, \text{ISQ}_9, \text{ISQ}_{10}) = (2^{-10}, 2^{-9}, \dots, 2^{-1}, 2^0, 2^1, \dots, 2^{10})$$

Which in shorthand is written as

$ISQ_n = (2^n)_{n \in \mathbb{N} \text{ and } n \in [-10, 10]}$, or, even $n \in \mathbb{N}$ may be taken as understood, and then as

$ISQ_n = (2^n)_{n \in [-10, 10]}$

13.2.3 Some Relevant Notations and Definitions

- A sequence is generally denoted as (S_n) , representing the infinite sequence $(S_1, S_2, S_3, \dots)$. However, for the *restricted* sequences, we mention the range of the integers which are used as indices of the restricted sequence. For example, $(S_n)_{n \in [-10, 10]}$ represents the sequence $(S_{-10}, S_{-9}, \dots, S_9, S_{10})$, and $(S_n)_{n \in [10, \infty]}$ represents the infinite sequence $(S_{10}, S_{11}, S_{12}, \dots)$.
- The constituents of a sequence are called *elements* or *terms* of the sequence. For example, the elements or terms of the sequence $(2, 3, 4, 2, 1, 1)$ are 1, 2, 3 and 4.
- The *domain* S for functional representation of a sequence $f: S \rightarrow \mathbb{R}$, is called *indexing set*. From the above examples, indexing set may be $[-10, 10]$.

The term *index* or *rank of an element* in a sequence is, *unfortunately not standard*. For our purpose, for the sequence $f: S \rightarrow \mathbb{R}$, *index* of an element, say r in the sequence is the element s of the domain S which maps to r . More explicitly, index of r of $\mathbb{R}$ is s such that $f(s) = r$.

For example, for sequence $ISQ_n = (2^n)_{n \in [-10, 10]}$, *index of* 2^{-10} is (-10) , because $ISQ(-10) = 2^{-10}$, though it is the first element in the sequence. Similarly, Index of $2=2^1$ is 1, because $ISQ(10) = 2^1$, though it is 12th in the sequence.

- **Length of a sequence:**

The *number of occurrences* of the terms in a sequence is called its length. For example, the length of the sequence $(1, 0, 1, 0, 1, \dots)$ is *infinite*. However, the corresponding set $\{0, 1\}$ has *only two elements*. From the above, it is clear that a sequence may be *finite* or *infinite*. A finite sequence having n elements is also called *n-tuple*.

- **Subsequence of a sequence:**

For a given sequence, say (S_n) , a sequence is called its *subsequence*, if

- The subsequence is obtained by deleting some of the terms of (S_n) , i. e. each term of subsequence occurs somewhere in (S_n) , and
- without disturbing the relative positions of the remaining elements, i. e.
 - if S_p and S_q are two terms of the sequence which are also in the subsequence, *and*
 - S_p precedes S_q in the sequence (S_n) , then S_p also precedes S_q in the subsequence.

For example, $(2, 4, 6, \dots)$ is a subsequence of $(1, 2, 3, \dots)$. **But** $(4, 2, 6, \dots)$ is **not** a subsequence of $(1, 2, 3, \dots)$.

- **Recursively definable sequences:** A sequence (a_n) is said to be recursively definable if the n^{th} term of the sequence can be obtained through a formula involving some or all the previous terms.

For example, we discussed earlier the sequence $(s_n) = (2^n)_{n \in \mathbb{N}}$.

This sequence *can be defined recursively* as: $s_0 = 2^0 = 1$; and $s_n = 2 \times s_{n-1}$, because

$$s_n = 2^n = 2 \times 2^{n-1} = 2 \times s_{n-1}.$$

Next, we discuss some of the well-known types of sequences, which are all recursively definable. In the rest of the discussion in the unit, we will assume that, unless it is mentioned otherwise, our sequences (S_n) are indexed by $n \in \mathbb{N}$, where 0 may be considered as belonging to $\mathbb{N}$.

Check Your Progress 1

- 1) Tell which of the following is a subsequence of $S = (4, 6, 8, \dots)$

a) $(6, 12, 16)$ b) $(12, 6, 16)$ c) $(8, 64, 243, \dots)$

.....

- 2) Write first five terms of each of the following, considering S_1 as first term

a) $S_n = [n + (-1)^n] / n$ b) $[1 - (1/(n+1))]^{1/(n+1)}$

.....

- 3) Write general term for the following sequence

a) $11/2, 11/8, 11/14, \dots$ b) $-1, 1, 5, 13, \dots$

.....

- 4) Define the following sequence *recursively*

a) $1, 4, 13, 40, \dots$ (b) $4, 9, 16, 25$

.....

13.2.4 Well-known Types of Sequences

Here, we consider some well-known sequences like; Arithmetic progression (A.P); Geometric progression (G.P); Harmonic Progression (H.P); Fibonacci sequence. All these sequences are recursively definable.

A) Arithmetic progression (A.P)

For some constant real numbers a and d , the sequence

$$a, a+d, a+2d, a+3d, \dots$$

is called **arithmetic sequence** or **arithmetic progression**. For this sequence, the number a is called the **initial term** and the number d is called the **common difference**.

n^{th} term of A. P: It can be easily verified that n^{th} term t_n of A. P., viz, (t_n) —with **initial term as $t_1=a$** , and **common difference as d** —is given by $t_n = a + (n-1) \times d$.

Example 13.6

- The sequence 3, 7, 11, 15, ... is an A.P. with *initial term as 3*, and *common difference as 4*
- The Sequence 5, 3.5, 2, 0.5, -1, -2.5, ... is an A. P. with *initial term as 5*, and *common difference as (-1.5)*.

$$40^{\text{th}} \text{ term of sequence (i)} = (3) + (40 - 1) \times 4 = 159.$$

$$21^{\text{st}} \text{ term of sequence (ii)} = (5) + (21 - 1) (-1.5) = -25.$$

Sum of first n terms of an A.P

$$(t_1, t_2, \dots, t_n, \dots)$$

is given by $S_n = (n/2) (t_1 + t_n)$

The *proof* of the above formula for sum S_n follows from the following argument:

$$S_n = t_1 + t_2 + \dots + t_{n-1} + t_n$$

Writing in reverse order, we get

$$S_n = t_n + t_{n-1} + t_2 + t_1$$

Summing corresponding terms, we get

$$2 S_n = (t_1 + t_n) + (t_2 + t_{n-1}) + \dots + (t_n + t_1)$$

But each of the sum of pair of terms within one pair of parentheses is $= (t_1 + t_n)$.

$$\text{For example, } (t_2 + t_{n-1}) = [(t_1 + d) + (t_n - d)] = (t_1 + t_n)$$

$$\text{Thus, } 2 S_n = n (t_1 + t_n) \quad \text{or} \quad S_n = (n/2) (t_1 + t_n).$$

For an A.P with initial term $t_1=a$ and common difference $= d$, the sum formula becomes

$$S_n = (n/2) (t_1 + t_n) = (n/2) [a + (a + (n-1) d)]$$

$$= (n/2) [2a + (n-1) d]. \dots \dots \dots (I)$$